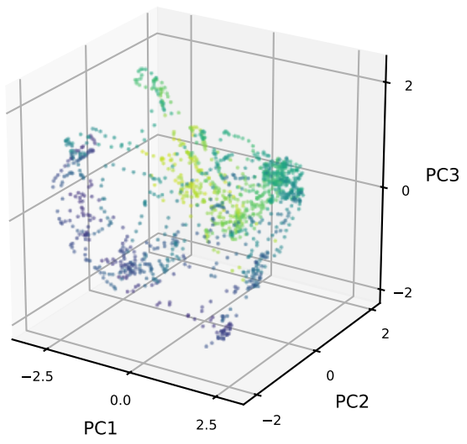
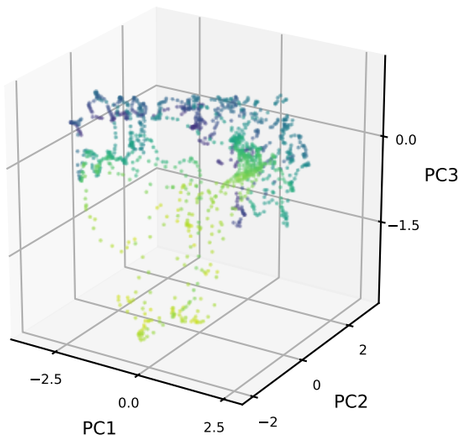
Frozen LSTMs at 1B frames · all seed pairs
Matched trajectory, fixed goal memory, eastward motion

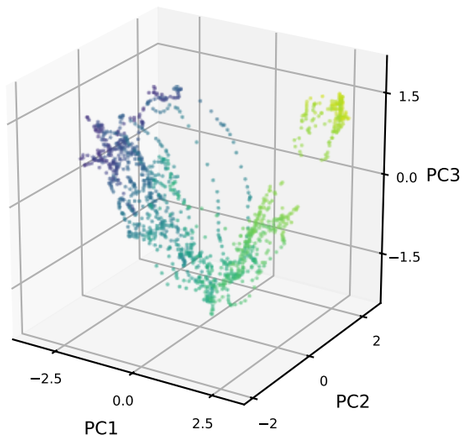
Seed 1 · Grid only
PC1–3: 54.2% of variance



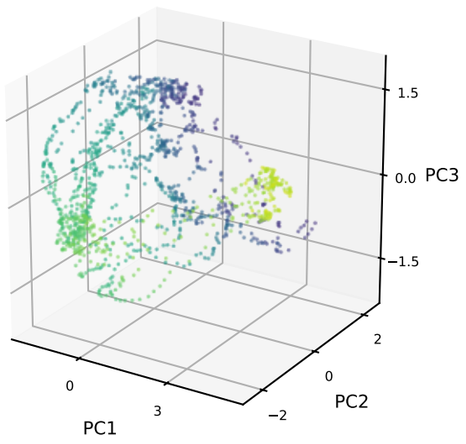
Seed 1 · Grid + 33% PGC
PC1–3: 56.2% of variance



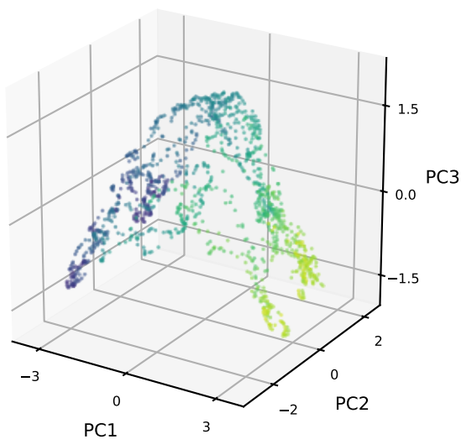
Seed 2 · Grid only
PC1–3: 60.9% of variance



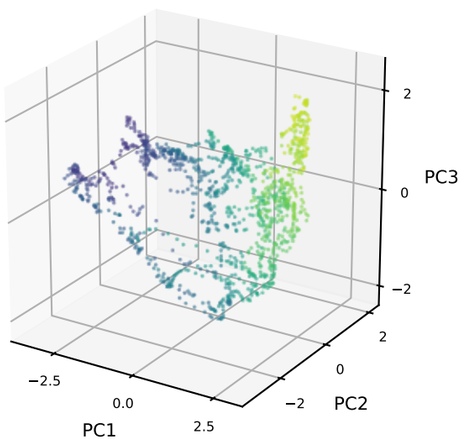
Seed 2 · Grid + 33% PGC
PC1–3: 62.5% of variance



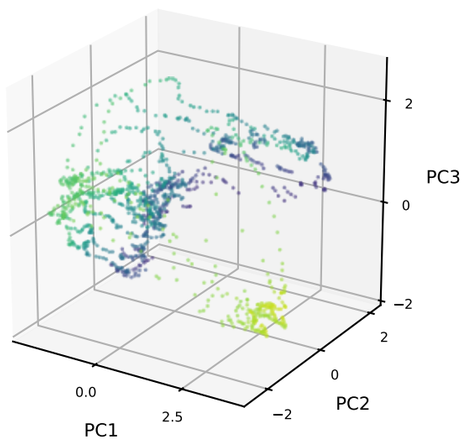
Seed 3 · Grid only
PC1–3: 60.5% of variance



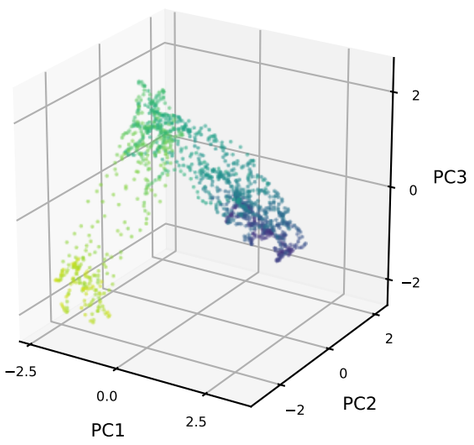
Seed 3 · Grid + 33% PGC
PC1–3: 58.1% of variance

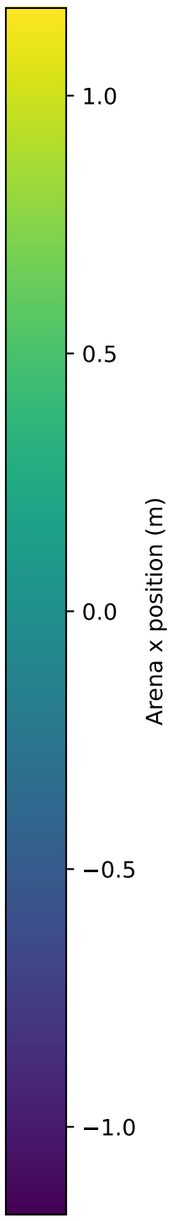


Seed 4 · Grid only
PC1–3: 60.3% of variance

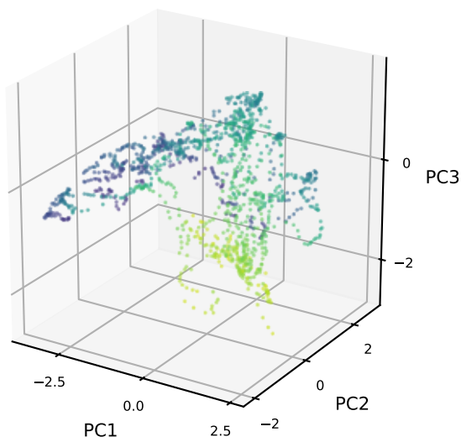


Seed 4 · Grid + 33% PGC
PC1–3: 54.7% of variance

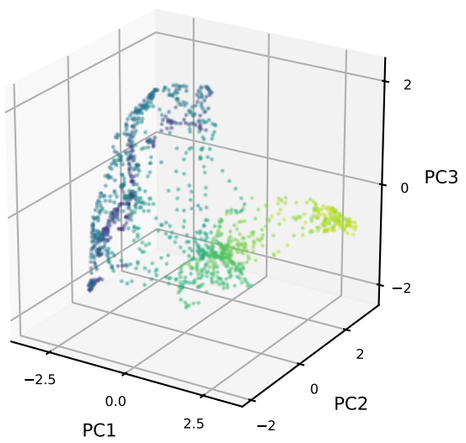




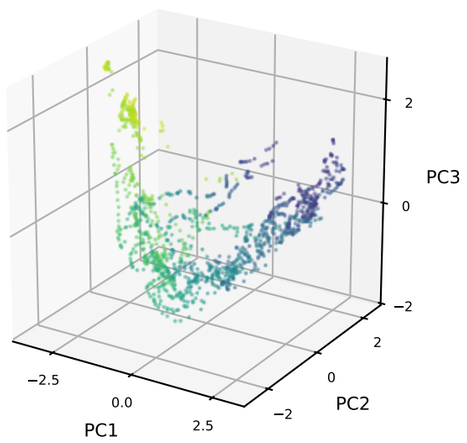
Seed 5 · Grid only
PC1–3: 50.2% of variance



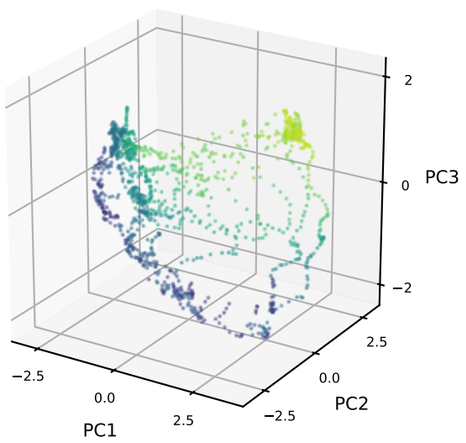
Seed 5 · Grid + 33% PGC
PC1–3: 63.5% of variance



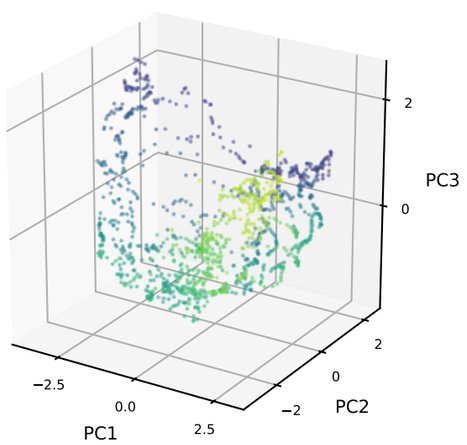
Seed 6 · Grid only
PC1–3: 55.2% of variance



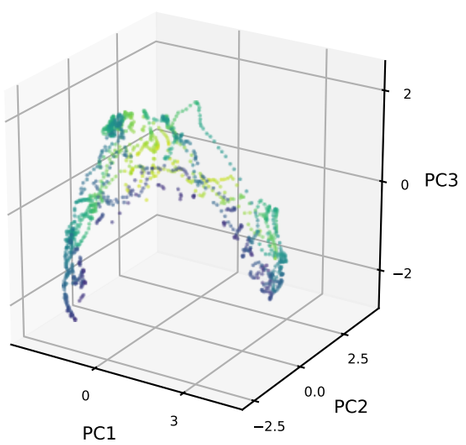
Seed 6 · Grid + 33% PGC
PC1–3: 61.2% of variance



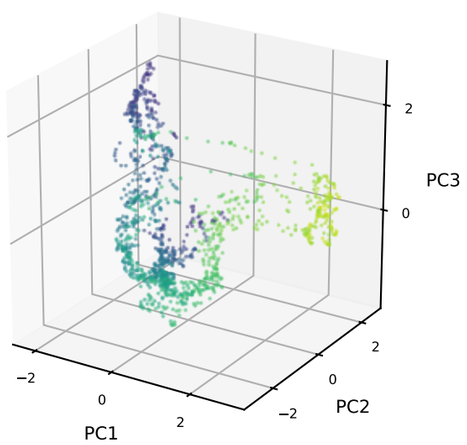
Seed 7 · Grid only
PC1–3: 57.1% of variance



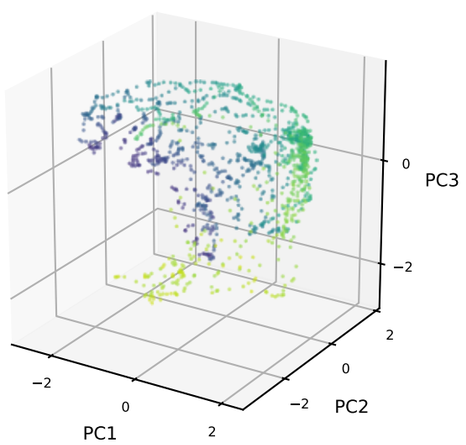
Seed 7 · Grid + 33% PGC
PC1–3: 63.5% of variance



Seed 8 · Grid only
PC1–3: 53.6% of variance



Seed 8 · Grid + 33% PGC
PC1–3: 57.4% of variance



Independent, centered PCA per model. PC axes are not aligned across models. No torus inference from these projections alone.